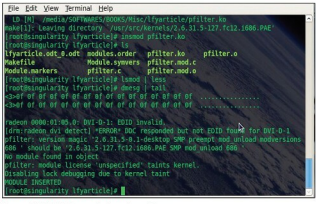

Figure 1: Module inserted after the make run

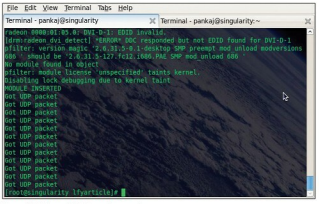

Figure 2: The module is getting packets

```
    nf_unregister_hook(&netfilter_ops_in); /*unregister NF_IP_PRE_ROUTING hook*/
    nf_unregister_hook(&netfilter_ops_out); /*unregister NF_IP_POST_
ROUTING hook*/

    printk(KERN_INFO "MODULE REMOVED\n");
}
```

In the above code, we register the *main_hook()* function to be called at the INET_PRE_ROUTING (for packets coming to the machine) and INET_POST_ROUTING (for packets leaving the machine) stages in Netfilter traversal (points [1] and [4] in the schematic diagram). We do this by assigning *main_hook()* as the *hook* member of *nf_hook_ops*. Next, we specify that the protocol family here is IP version 4 by assigning PF_INET to the *pf* member of *nf_hook_ops*. (For IP version 6, it would be PF_INET6.) Next comes the priority: the packet-filter functions within the rules list of a hook are sorted by the priority field in ascending order, and will be invoked in this order when a packet transits. (The priority values are defined in *linux/netfilter_ipv4.h*.) Here, we have assigned the first priority for our filter.

Now we are ready to compile the module; let's create a *makefile*, for convenience:

```
obj-m += pfilter.o
all:
    make -C /lib/modules/$(shell uname -r)/build M=$(PWD) modules
clean:
    make -C /lib/modules/$(shell uname -r)/build M=$(PWD) clean
```

After the *make* run in the directory containing the code, we'll have a number of output files, including the loadable module *pfilter.ko*. We insert the module using the command *insmod*, as the root user.

```
# insmod pfilter.ko
```

List the inserted modules by running *lsmod*. You can see the 'MODULE INSERTED' output from *printk()* by running the following command:

```
# dmesg | tail
```

Figure 1 shows the output in the terminal window just after loading the module, and Figure 2 shows the output after the filter has received packets. By the way, since DNS lookups are done via UDP, DNS won't work on your system while this module is loaded. To get back to normal, just remove the module by running *rmmod pfilter*. The cleanup code will un-register the hooks, and DNS will work again. Indulge in Netfilter and see its power. FOSS ROCKS!! END

References

netfilter.org
vger.kernel.org/~davem/skb.html
wikipedia.org (Check out the protocols, and visit ietf.org with the RFC number)
The LFY column series, A Voyage to the Kernel, will be helpful in gaining kernel-related information.

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